
FACULTY OF SOFTWARE ENGINEERING

DELIVERY AUTO ALERT MANAGEMENT SYSTEM

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SOFTWARE ENGINEERING PROJECT REPORT

Java Servlet & JSP · MVC Architecture · SQL Server 2019

NetBeans 13 · Apache Tomcat 9 · Maven

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1. Introduction

The **Delivery Auto Alert Management System** is a web-based application designed to manage delivery operations, warehouse inventory, and automatic alert notifications. The system helps businesses track orders, shipments, and stock movement while also generating alerts when specific conditions occur.

This system is developed using **Java Servlet and JSP** with NetBeans IDE, following the **MVC (Model – View – Controller)** architecture. The application allows administrators to manage products, customers, warehouses, deliveries, and payments, while helping users monitor delivery status and related information.

The main objective of this project is to improve delivery management efficiency and reduce manual monitoring by automatically generating alerts when operational events occur.

2. Software Requirements

The system requires the following software, tools, and configurations for development and deployment.

2.1 Development Environment

Category	Technology	Details
IDE	NetBeans IDE 13	Primary development environment
Runtime	Java JDK 8 (javax.*)	javax.servlet — Tomcat 9 compatible only
Server	Apache Tomcat 9	Web application server, port 8080
Build Tool	Maven	Dependency management and WAR packaging
Database	SQL Server 2019	localhost:1433, DB: DeliveryAutoAlert
JDBC Driver	sqljdbc4.jar (v4.0)	Microsoft SQL Server JDBC driver
Frontend	HTML / CSS / JSP	Inter font, responsive layout, MVC2 views
Architecture	MVC2 Pattern	Servlets as controllers, JSPs as views

2.2 Database

- Microsoft SQL Server 2019 running locally on port **1433**
- Database name: **DeliveryAutoAlert**
- Schema file: **DeliveryAutoAlert_Schema.sql** (creates all 19 tables and seed data)
- Default admin login: username **admin** / password **admin123**
- Passwords stored as **SHA-256** hex strings in the Users table

2.3 System Architecture (MVC2)

Layer	Components	Responsibility
Model	DTO, DAO classes	Data transfer objects and JDBC database queries
View	JSP pages (33)	HTML rendering and form submission to controllers
Controller	Servlets (14)	HTTP request handling and application logic
Filter	AuthFilter	Session validation and access control on all routes
Utils	DBUtils.java	JDBC connection factory (Class.forName + DriverManager)

3. Diagrams

3.1 Entity Relationship Diagram (ERD)

The ERD below shows all tables in the DeliveryAutoAlert database and the relationships between them, generated directly from SQL Server Management Studio. The diagram is displayed on a landscape page for readability.

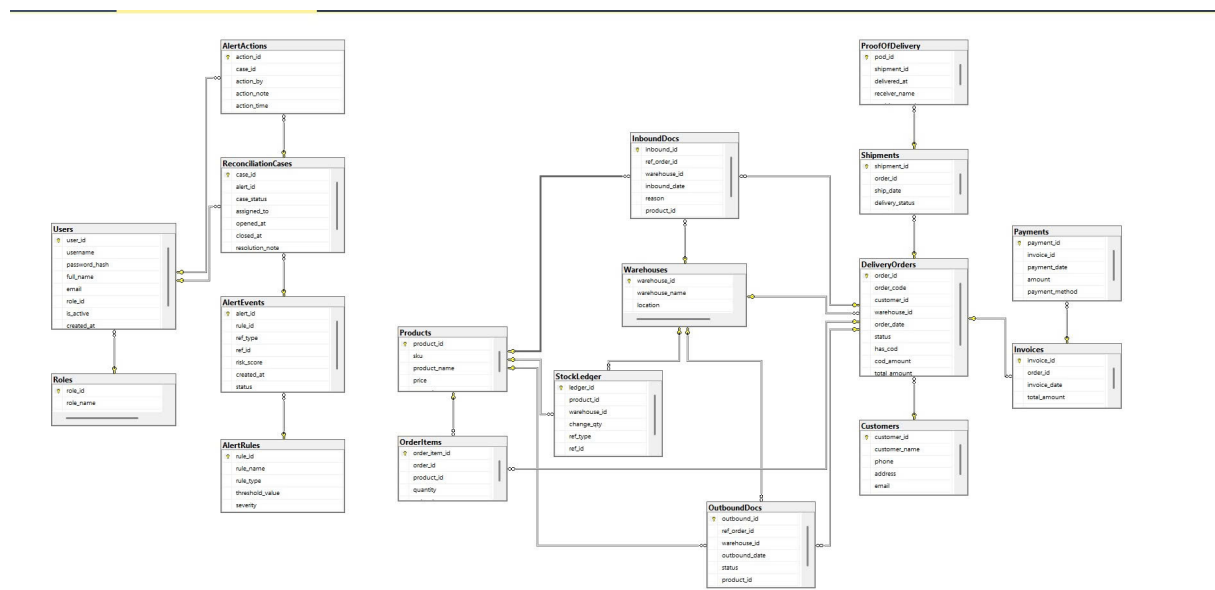


Figure 3.1 — Entity Relationship Diagram — DeliveryAutoAlert Database Schema

Key relationships shown in the diagram:

- A **Customer** can have multiple **DeliveryOrders**
- A **DeliveryOrder** contains multiple **OrderItems** (product lines)
- A **Shipment** is dispatched per **DeliveryOrder**, confirmed by a **ProofOfDelivery**
- An **Invoice** and **Payment** are generated per **DeliveryOrder**
- Stock movements are recorded in **StockLedger** per warehouse and product
- An **AlertEvent** is fired by an **AlertRule** and opens a **ReconciliationCase**
- **AlertActions** log all steps taken to resolve a **ReconciliationCase**

3.2 Use Case Diagram

The Use Case Diagram below shows how the two main system actors — **Customer** and **Admin** — interact with the system. Use cases are grouped by functional module.

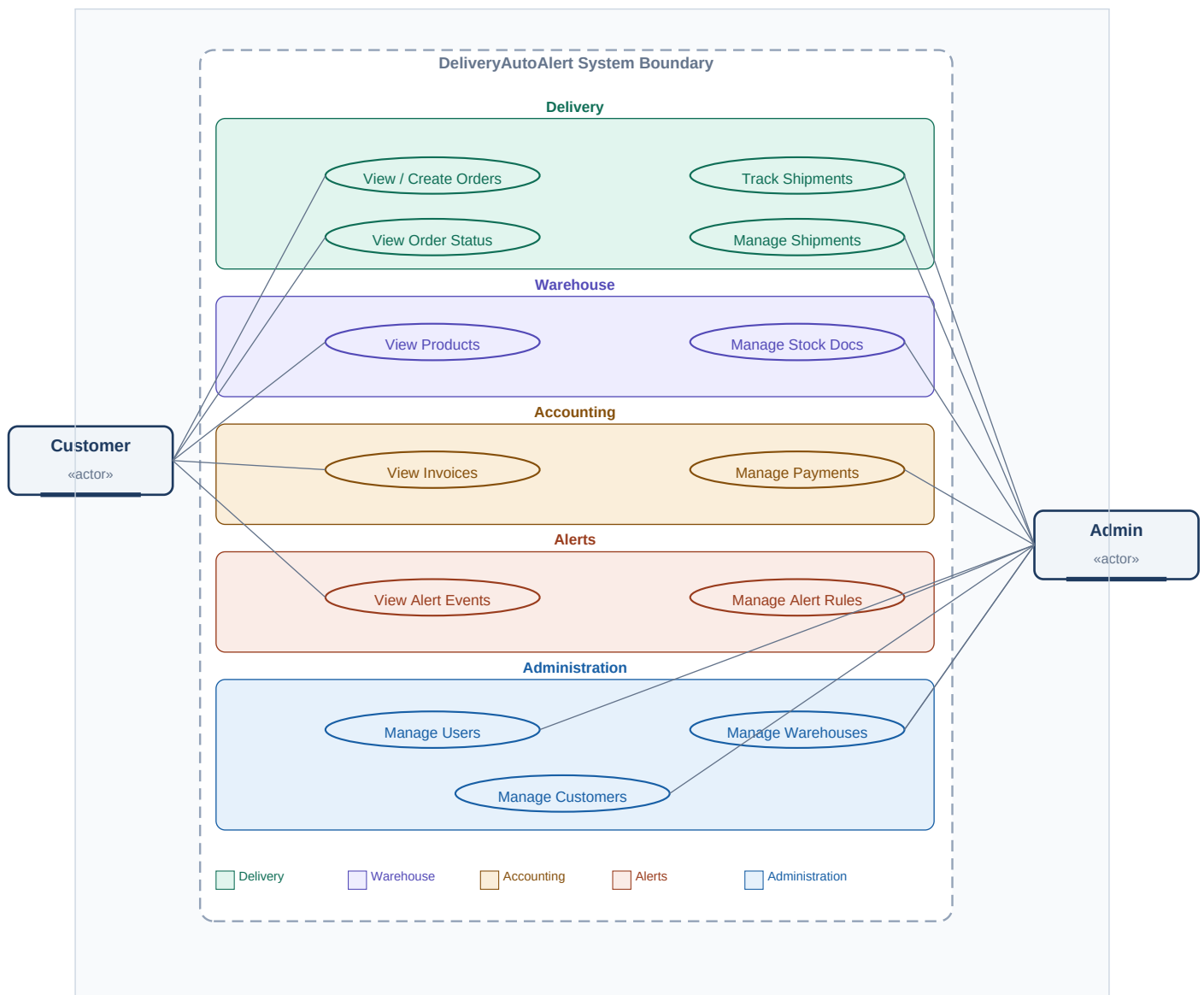


Figure 3.2 — Use Case Diagram — DeliveryAutoAlert System

4. Functions

4.1 Customer Functions

Customers use the system to monitor their deliveries and related information:

View Products

Customers can view a list of available products with name, SKU, price, and details.

Create Delivery Order

Customers can submit a delivery order containing product information, quantities, and delivery destination.

View Order Status

Customers can check the current status of their orders: Pending, Dispatched, Delivered, Returned, or Cancelled.

Track Shipment

Customers can view shipment details including dispatch date, route, delivery status, and proof of delivery.

View Invoice

Customers can view the invoice generated for their order, including the total amount and current payment status.

4.2 Admin Functions

The administrator has full control of the system and manages all operational data across the entire delivery lifecycle.

Customer Management

Admins can create, update, or delete customer records including contact details and address information.

Product Management

Admins manage the product catalog including SKU, product name, price, and availability per warehouse.

Warehouse Management

Admins manage warehouse locations and track inventory. Each warehouse maintains its own stock ledger and product catalog.

Delivery Order Management

Admins can view and manage all delivery orders including status tracking, COD amounts, and order item details.

Shipment Management

Admins create and update shipment records including dispatch details, delivery routes, and proof of delivery uploads.

User Management

Admins manage system users and assign roles (Admin, Delivery, Accounting, Warehouse) to control access permissions.

Alert Rule Management

Admins configure alert rules with thresholds and severity levels. Rules fire for: late deliveries, quantity mismatches, COD discrepancies, missing proof of delivery, and negative stock.

Alert Monitoring

Admins view alert events generated by the system and investigate reconciliation cases to resolve detected issues.

Invoice and Payment Management

Admins manage invoices linked to delivery orders and record payment transactions including method and status tracking.

5. References

The following resources were referenced during the development of this project:

- [1] **Oracle Java Documentation**
<https://docs.oracle.com/javase/>
- [2] **Apache Tomcat Documentation**
<https://tomcat.apache.org/>
- [3] **JDBC API Documentation**
<https://docs.oracle.com/javase/tutorial/jdbc/>
- [4] **NetBeans IDE Documentation**
<https://netbeans.apache.org/>
- [5] **UML Diagram Tools — draw.io**
<https://app.diagrams.net/>
- [6] **Maven — Build Tool Documentation**
<https://maven.apache.org/guides/>
- [7] **SQL Server 2019 Documentation**
<https://learn.microsoft.com/en-us/sql/sql-server/>